

Risk-taking incentives predict aggression heuristics in female gorillas

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This **important** study uses long-term behavioural observations to understand the factors that influence female-on-female aggression in gorilla social groups. The evidence supporting the claims is **convincing**, as it includes novel methods of assessing aggression and considers other potential factors. The work will be of interest to broad biologists working on the social interactions of animals.

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Abstract

Competition is commonly reflected in aggressive interactions among groupmates, as individuals try to attain or maintain higher social ranks that can offer them better access to critical resources. In this study, we investigate the factors that can shift competitive incentives against higher- or lower-ranking groupmates, that is, more or less powerful individuals. We use a long-term behavioural dataset on five wild groups of the two gorilla species starting in 1998, and we show that most aggression is directed from higher- to lower-ranking adult females close in rank, highlighting rank-reinforcement incentives. Yet, females directed 42% of aggression to higher-ranking females than themselves. Females targeted groupmates of higher rank with increasing number of males in the group, suggesting that males might buffer female-female aggression risk. Contrarily, they targeted females of lower rank with increasing number of females in the group, potentially because this is a low risk option that females prefer when they have access to a larger pool of competitors to choose from. Lactating and pregnant females, especially those in the latest stage of pregnancy, targeted groupmates of higher rank than the groupmates that cycling females targeted, suggesting that energetic needs may motivate females to risk confrontation with more powerful rivals. Our study provides critical insights into the evolution of competitive behaviour, showing that aggression heuristics, the simple rules that animals use to guide their aggressive interactions, are not simply species-specific but also dependent on the conditions that populations and individuals experience.

Introduction

Animals that live in groups often compete for access to resources such as food and mates [1, 2]. The potential costs of this competition can drive the formation of hierarchies that determine priority of access to resources without superfluous conflicts [3, 4]. Accordingly, individuals may choose strategically who they compete with, in order to minimize costs and maximize gains. Previous research suggests that individuals primarily compete with those closest in the hierarchy, to attain (against close higher-ranking) or maintain (against close lower-ranking individuals) their ranks ('rank-dependent aggression'; [5–10]). Aggression, probably the most straightforward proxy of competition, commonly increases in frequency when resource availability is lower and it is most often directed towards lower-ranking individuals [8, 11]. Yet, recent research has demonstrated that the rules that animals use to guide their aggressive interactions ('aggression heuristics') towards groupmates of different ranks vary even within species [6, 8]. In this study, we test the hypothesis that this variation arises due to the different conditions that (individual) animals experience, and specifically that the social environment and individual energetic needs shape competitive incentives and aggression towards individuals of different ranks.

Regarding the social environment, larger group sizes may entail a larger number of low-ranking individuals which are cumulatively targeted, as competition and aggression are preferentially directed towards the lowest-ranking groupmates (see also 'bullying' in [8]). When individuals have access to a larger pool of competitors, they may preferentially target less powerful ones, consistent with risk-sensitivity theory, which posits that individuals tend to minimize risks when they have the option [12, 13]. Conversely, larger group sizes might entail a larger number of allies (e.g., kin for spotted hyaenas, *Crocuta crocuta*, [14]) or protectors (e.g., adult males for female gorillas [11, 15]), which can increase social support and, eventually, minimize any risk-related costs of aggression towards higher-ranking competitors. Hence, group size, the overall number of individuals in the group, might be a poor predictor of aggression because it conflates opposing effects of different kinds of individuals (e.g., see [11] for an example of opposing effects of the number of females and number of males on female gorilla aggression). Finally, if larger group sizes promote within group competition increasing overall aggression rates [2, 11, 16], they might simultaneously promote high-ranking individuals to reinforce their status by targeting lower-ranking groupmates and lower-ranking individuals to direct aggression against higher-ranking groupmates if this can allow them access to precious resources.

Regarding the energetic needs, greater needs for resources may boost the incentives of higher-ranking individuals to reinforce their status by strategically directing their competitive efforts towards lower-ranking groupmates [8, 9, 17]. Contrarily, greater individual needs may also prompt low-ranking individuals, who struggle more to access resources and experience a lower risk-reward ratio (more to gain, less to lose), to show greater aggression rates even against higher-ranking groupmates [18–20]. This aggression may help individuals to improve their ranks but it might be risky if it can incite retaliation from high-ranking, powerful, recipients. Yet, the benefits related to status improvement (long-term benefit) or resource acquisition (short-term benefit) might counterbalance any risk-related costs. Various empirical examples support the latter hypothesis: hunger-driven payoff asymmetries can increase aggression from lower- to higher-ranking individuals (noble crayfish, *Astacus astacus*; [21]), reproductive suppression of low-ranking females can promote conflict escalation against high-ranking ones (paper wasps, *Polistes dominulus*; [22]) and the energetic/nutritional needs of pregnancy (due to support of fetal growth) or lactation (due to milk production) can increase female aggression that reverses hierarchical relationships [23–25].

Studies investigating the influence of social, ecological or physiological factors on aggression patterns across species usually focus on aggression frequency/rate [26, 27]; in this study, we build on this literature to test how relevant factors influence ‘aggression direction’ in terms of power differentials, aiming to unravel another evolutionary aspect of competitive strategies. Gorillas represent an intriguing case for this endeavour, because females of both species form surprisingly stable hierarchical relationships, usually maintained over their whole co-residence in a group, but they often direct aggression towards higher-ranking groupmates [28–30]. Female-female aggression rates towards both higher- and lower-ranking rivals decrease with the number of adult males/protectors in the group but increase with the number of females/competitors in the group [11, 29–31]. Additionally, aggression rates are greater in pregnant females [11] which, like lactating females, spend more time feeding [32], highlighting the greater energetic needs of females in these reproductive states – similar to humans and other apes [33].

We use behavioural observations on one wild western (*Gorilla gorilla gorilla*) and four wild mountain (*Gorilla beringei beringei*) gorilla groups, starting in 1998 in one of the mountain gorilla groups, to test if the social environment or energetic needs influence female aggression towards more or less powerful females. Specifically, we test whether females direct aggression of lower or higher score (score = recipient-aggressor rank difference) depending on (i) the number of males in their group who may support or protect females, (ii) the number of females in the group representing competitors over resources, and (iii) female reproductive state (cycling, stage of pregnancy, or lactation) which is an (indirect) proxy of energetic needs. We compare our results on the direction of aggression to previous results examining the effects of the same variables on aggression frequency/rates.

Methods

Study system and behavioural data

We studied one western gorilla group (ATA/Atanangad; **Table 1**) in Loango National Park, Gabon, and four mountain gorilla groups in Bwindi Impenetrable National Park, Uganda (**Table 1**). Observations of western gorillas lasted typically between 07:00 and 16:30h but observations of mountain gorillas were limited to 4 hours per day, typically between 08:00 and 15:00h following the regulations of the Uganda Wildlife Authority.

Trained observers recorded both focal and ad libitum behavioural observations, including decided avoidance (when an individual walks away from another approaching individual) and displacement (when an individual avoids another and the latter takes the place of the first) behaviours. These two behaviours are ritualized, occurring in absence of aggression, they are considered a more reliable proxy of power relationships over aggression [34], and they are typically used to infer gorilla hierarchical relationships [29, 30]. Similar to the recent studies which have inferred social ranks in gorillas [11, 35], we used all avoidance and displacement interactions throughout the study period and we used the function *elo.seq* from R package *EloRating* [36] to infer daily individual female Elo-scores [37, 38]. We assigned to all avoidance/displacement interactions equal intensity, that is, equal influence to the power relationship of the interacting individuals ($k=100$). We also assigned to individuals present at the onset of the study initial Elo-scores of 1000 and to individuals entering the hierarchy later (e.g. maturing individuals or immigrants) the score of the lowest ranking individual during the entrance day [11]. This method takes into account the temporal sequence of interactions and updates an individual’s Elo-scores each day the individual interacted with another. The Elo-scores of the winner and loser of an interaction are updated as a function of the winning probabilities prior the interaction: winners with low winning probabilities get greater score increases than winners with high winning probabilities and losers with low winning probabilities get smaller

Group	Period	# of females	Aggression
ATA	2017-2023	3.30±0.85 (6)	208 / 65 / 27
BIT	2015-2023	4.14±0.35 (6)	509 / 35 / 23
KYA	1999-2016	6.07±1.02 (9)	3256 / 445 / 315
MUK	2016-2023	6.59±1.20 (8)	1389 / 89 / 141
ORU	2014-2023	5.59±0.96 (8)	355 / 7 / 7

Table 1

Study groups, study period, average number of females per day ($\pm$ s.d.; total number in parentheses), and number of aggressive interactions (mild/moderate/severe).

score decreases than losers with high winning probabilities. We present these interactions and hierarchies in detail in [28] (see “traditional Elo rating method”; we do not use the “optimized Elo-rating method” as it yields similar results and it is not widely used). We standardized Elo-scores per group and day such that the highest-score was 1 and the lowest 0.

The observers also recorded aggressive behaviours among adult females (>10 years old for western; >8 years old for mountain female gorillas [11]), which can be context dependent [34]. For our analysis, we classified these behaviours into three intensity categories (as per [30, 39]): mild (cough/pig grunt [aggressive vocalization that sounds like a pig grunting or a deep throated cough], soft bark [characteristic aggressive vocalization], bark, scream and pull vegetation), moderate (chest-beat [beating of chest by hands], strut-stand/run [stand in front of or runs past another with a distinctive stiff leg stance], lunge [aggressively and quickly leaning forward towards another], direct charge, indirect charge, run at/over, push) and severe aggression (hit [aggressively strike/slap/push, etc another], attack [jumps on, bites, etc. another], drag [pulls another along the ground], fight [two individuals attack, bite, scream etc. at each other], bite, chase, kick). To quantify “aggression direction”, we assigned a score to each aggressive interaction, calculated by subtracting the standardized Elo-score of the aggressor from that of the recipient. The score was maximum (=1) for interactions where the aggressor was the lowest-ranking female and the recipient was the highest-ranking female; and minimum (=−1) for interactions where the aggressor was the highest-ranking female and the recipient was the lowest-ranking female. More generally, positive scores represented aggression up and negative scores represented aggression down the hierarchy (Figure 1).

We used demographic data to estimate daily female reproductive state. On a given day, we classified as ‘pregnant’ any female that gave birth 255 days or less after that day [40], as ‘cycling’ any female that was not classified as pregnant and she had been observed mating since her last parturition and as ‘lactating’ any female with a dependent infant (based on last observation of nipple contact; [41, 42]) that was not pregnant and had not observed mating since her last parturition. Lactation is often considered more energetically demanding than pregnancy as a whole but the latest stages of pregnancy are highly energetically demanding, potentially even more than lactation [43, 44]. Thus, we differentiated between the first (1–85th day of pregnancy), second (85–170th day), and third (170–255th day) trimester of pregnancy (85 days each).

Statistical analyses

We fitted a linear mixed effects model with a logit function to test whether females who have greater energetic needs and/or experience different social environments, direct aggression of higher or lower score (response variable, continuous, between −1 and 1; Figure 1) to other females. Given that we tested for the aggression direction and not aggression rates, the design of our analysis was independent of observation effort, and thus, we were able to use both focal and ad libitum observations. Specifically, we considered each aggressive interaction recorded during either a focal or an libitum observation as a separate data point. In our model, we fitted the following explanatory variables: aggression intensity to test if aggression of greater score is more often mild than moderate or severe; number of adult males in the group (>14 years old for western males; >12 years old for mountain males); number of females in the group; reproductive state of the aggressor (cycling, trimester of pregnancy, or lactation); and species (western or mountain). We fitted the identities of interacting females, dyad and group as random factors. Finally, we used Tukey post hoc comparisons (via the *glht* function from the *multcomp* package [45]), to perform pairwise comparisons between all reproductive states.

We ran the model in R version 4.1.2 using the function *glmmTMB* from the package *glmmTMB* [46]. We used the function *Anova* from package *car* [47] to test the significance of fixed factors and to compute 95% confidence intervals. We tested the residual distributions, using the functions *testDispersion* and *testUniformity* from package *DHARMa* version 0.4.6 [48] to validate

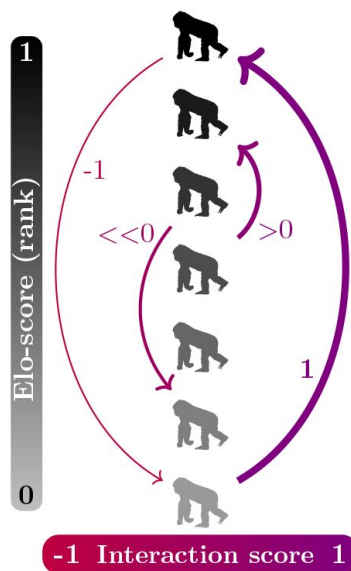


Figure 1.

Calculation of the interaction score.

The lower the rank of the aggressor and the greater the rank of the recipient, the greater the score (-1 to 1; line thickness). Arrows start from aggressors and point to recipients of aggression. Figure created using a female gorilla silhouette icon from <http://phylopic.org> (created by T. Michael Keesey) and TikZ (TeX).

the model. We used the base function *cor.test* and the function *check collinearity* from the *performance* package to test for correlations and multicollinearities of the explanatory variables: all VIF (variance inflation factor) values were <1.5 indicating no serious multicollinearities [49].

Results

We analyzed 6871 aggressive interactions among a total of 31 adult female gorillas in the five social groups (Table 2). The average percentage of aggressive interactions directed from lower- to higher-ranking females across groups was $41.8 \pm 6.7\%$ ($\pm$ SD; ATA: 47.2%; BIT: 46.3%; KYA: 36.5%; MUK: 46.1%; ORU: 32.7%). For comparison, only $16.4 \pm 4.1\%$ of displacement/avoidance interactions that we used to infer the highly stable hierarchies (details in [28]) were directed from lower- to higher-ranking females ($\pm$ SD; ATA: 15.4%; BIT: 19.2%; KYA: 12.8%; MUK: 22.1%; ORU: 12.5%). This result confirms previous evidence that aggression may not be a reliable proxy of power in gorillas or other species [29, 30, 34].

Aggression from lower- to higher-ranking females was 85% mild, 9% moderate and 6% severe while aggression from higher- to lower-ranking females was 82% mild, 10% moderate and 8% severe. Generally, aggression of different intensity showed similar distributions and all aggression was most common from higher- to lower-ranking females close in rank ($-0.5 > \text{score} > 0$; Figure 2). The interactions of mild aggression were of greater score (recipient aggressor rank difference) than interactions of moderate aggression, meaning that females were more likely to use mild rather than moderate aggression against more powerful rivals, but the score difference between interactions of severe and moderate or mild aggression was not significant (Table 2).

Female gorillas directed aggression of greater score when there were more males in the group and when there were fewer females in the group (Figure 3; Table 2). When we ran our analysis testing for group size (number of weaned individuals in the group), instead of the numbers of males and females, its influence on interaction score was not significant (estimate = -0.001, p-value = 0.682). Females in the third trimester of pregnancy directed the aggression of the highest score, females in any pregnancy stage directed aggression of greater score than lactating and cycling females, and lactating females directed aggression of greater score than cycling females, likely highlighting an effect of energetic needs on aggression heuristics (Figure 3; Table 2). Post hoc pairwise comparisons showed significant differences between all reproductive states, apart from the difference of the first and the other two trimesters of pregnancy (Table 3; Figure 3). When we ran our analysis merging pregnancy trimesters into one category, we found pregnant females to direct aggression of significantly greater score than lactating and cycling females, and lactating females to direct aggression of significantly greater score than cycling females (not shown). Finally, our results did not show any significant difference between aggression score within the one western and four mountain gorillas groups.

Notably, a positive (or negative) correlation of a predictor with the interaction score does not necessarily represent a shift from aggression towards females lower-ranking than the aggressor to aggression towards females higher-ranking than the aggressor (or vice versa) – but, more generally, it represents a shift of aggression towards more (or less) powerful females independently of the rank relationship to the aggressor. For example, females in the second trimester of pregnancy direct most aggression towards rivals higher-ranking than themselves; yet they direct aggression of significantly lower score than females in the third trimester of pregnancy, that is, the latter direct more aggression to females even more higher-ranking than those during second trimester (Figure 3; Table 2).

Table 2

Results from the linear mixed effects model.

Significant p-values appear in bold. The significance of each level of a categorical variable was evaluated against the reference level (placed in parenthesis) according to whether their confidence intervals (CI) include zero or not. 'Pregnant n' denotes the nth trimester of pregnancy. To highlight that aggression rates can increase due to increase in interactions of different score, we also include the effect of some of the tested variables on overall adult female aggression rates, based on results of linear mixed effects models from [11] on the right of the table. 'ns': non-significant correlation; '+': positive correlation; '-': negative correlation; 'na': not tested (see [11] for details).

Response variable: Interaction score (recipient-aggressor rank difference)						Aggression
Fixed Factor	Level	Estimate	CI 95%	Chisq	P-value	Rate [11]
- Species (Mountain)	Western	0.088	[-0.495;0.671]	0.088	0.767	ns
- Reproductive state (Cycling)	Pregnant_1	0.130	[0.101;0.159]	138.812	< 0.001	+
	Pregnant_2	0.093	[0.063;0.123]	138.812	< 0.001	+
	Pregnant_3	0.146	[0.114;0.177]	138.812	< 0.001	+
	Lactating	0.043	[0.027;0.059]	138.812	< 0.001	-
- Number of females		-0.011	[-0.018;-0.003]	8.045	0.005	+
- Number of males		0.018	[0.010;0.026]	19.784	< 0.001	-
- Aggression intensity (Mild)	Moderate	-0.030	[-0.051;-0.009]	8.531	0.005	na
	Severe	-0.014	[-0.036;0.009]	8.531	0.229	na

* Pregnancy was not divided in trimesters in [11]

Figure 2.

Distribution of interaction score (recipient-aggressor rank difference): density of the mild, moderate and severe aggression as a function of the interaction score.

Positive scores represented aggression up and negative scores represented aggression down the hierarchy.

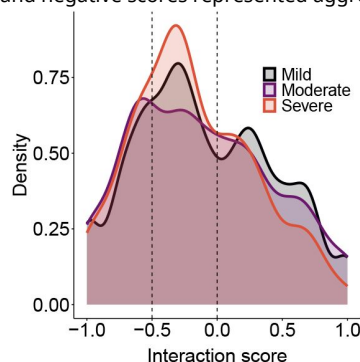


Table 3

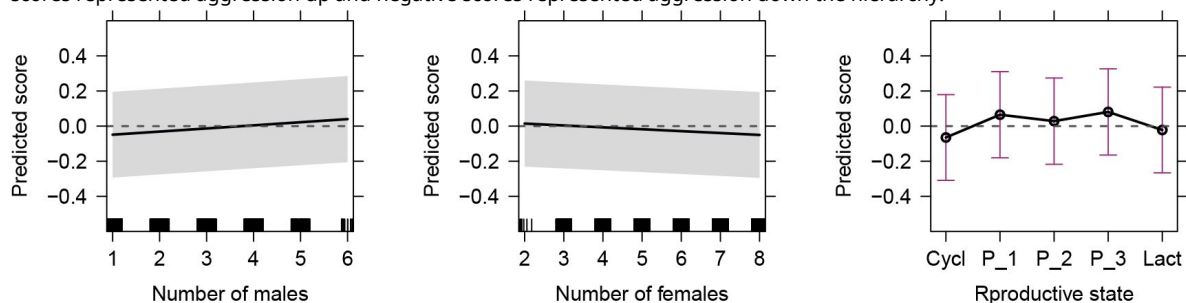
Post hoc comparisons of the different reproductive state. Significant p-values appear in bold. Cycl: cycling; P n: nth pregnancy trimester; Lact: lactating.

Comparison	Estimate	Std.Error	Z value	p value
P_1 - Cycl	0.130	0.015	8.681	0.000
P_2 - Cycl	0.093	0.015	6.071	0.000
P_3 - Cycl	0.146	0.016	9.147	0.000
Lact - Cycl	0.043	0.008	5.236	0.000
P_1 - P_2	-0.037	0.019	-1.965	0.267
P_1 - P_3	0.016	0.019	0.812	0.922
Lact - P_1	-0.087	0.015	-5.888	0.000
P_2 - P_3	0.052	0.019	2.721	0.047
Lact - P_2	-0.050	0.015	-3.368	0.006
Lact - P_3	-0.103	0.015	-6.715	0.000

Figure 3.

Predicted interaction score (recipient-aggressor rank difference) as a function of the explanatory variables of the linear mixed effects model, with a significant effect: number of adult males in the group, number of adult females in the group and aggressor's reproductive state (Cycl: cycling; P n: nth pregnancy trimester; Lact: lactating).

Shaded areas and whiskers show 95% confidence intervals. We created the figure using R package *effects* [50]. Positive scores represented aggression up and negative scores represented aggression down the hierarchy.



Discussion

Most aggression was directed from higher to lower-ranking females, usually close in rank, supporting the hypothesis that individuals commonly use aggression to reinforce their status. However, approximately 42% of aggression was directed from lower- to higher-ranking females, which is even greater than previous estimations in gorillas (34% [29] & 25% [30]) and also greater than in many other animals [8]. Our results suggest that such aggression towards more powerful rivals reflects competitive incentives influenced by the social environment and driven by circumstantial needs. This interpretation is in line with previous observations showing that female gorillas are usually unable to improve their ranks through active competition, as they form highly stable hierarchical relationships [28], that is, aggression is unlikely to be used for challenging the hierarchy. Importantly, female gorillas are more likely to respond aggressively ('retaliate') than submissively to aggression from other females [29], meaning that aggression involves some risk for the aggressor, especially when it targets more powerful groupmates: greater interaction score, greater risk (the greater the relative rank of the recipient, the greater the risk for the aggressor). Thus, our results further suggest that the social environment and circumstantial needs may influence individual decisions to engage in risky behaviours.

Female gorillas appeared to direct aggression of greater score when there were more males in the group, potentially supporting our interpretation that male support or protection [15, 29] provides females with an environment to take greater risks. Males may support lower-ranking females in order to decrease competitive inequities among females (e.g., to prevent low ranking female emigration from their group; [51]) and high-ranking females might hesitate to retaliate to aggressors and escalate a contest if more males are adjacent to the aggressors or are simply present in the group and can intervene. Interestingly, when females have access to fewer males, they generally exhibit greater aggression rates towards other females ([11]; **Table 2**, column 'Aggression rates'), potentially competing for male protection per se; but once they have access to more males (and male protection), they appear to direct more aggression to more powerful female rivals.

In contrast to the number of adult males, the number of adult females in the group was negatively correlated with interaction score, that is, female gorillas directed aggression to lower-ranking, less powerful, females when there were more females in the group. Hence, the previously observed increase in aggression rates in groups with more females ([11]; **Table 2**, column 'Aggression rates') likely pertains predominantly to aggression from more to less powerful rivals. Our result may reflect that females preferably target less powerful rivals when they have access to a larger pool of competitors to choose from, similar to humans ([52]; see also risk-sensitivity theory in introduction). Alternatively, the increased competition due to a larger number of competitors may prompt higher-ranking females to reinforce their status more than it prompts lower-ranking females to target higher-ranking ones in order to access resources. Overall, the combination of our present and previous results [11] showing the influence of the number of males and females on both female aggression rates and aggression direction confirms that non-human animals can adapt their aggression patterns according to the social context or the available social information (see also [53]).

Aggression score was also influenced by energetic needs: female gorillas in the most energetically demanding reproductive states directed aggression to more powerful females (greater aggression score/ recipient-aggressor rank difference). Females in the last and most energetically demanding stage of pregnancy directed aggression of greater score than all other females (this difference was significant for all reproductive states except females in the first trimester of pregnancy), females in any stage of pregnancy directed aggression of greater score than lactating and cycling females, and lactating females directed aggression of greater score than cycling females. While lactating

females can potentially have greater energetic needs than pregnant females, they might direct aggression of lower score than pregnant females because they show lower risk-tolerance in aggression towards higher-ranking groupmates in order to protect their dependent infants, reminiscent of risk-avoidance behaviours of lactating female African wild dogs (*Lycaon pictus*; [54]), and black bears (*Ursus americanus*; [55]). This interpretation is consistent with previous results showing that lactating females show the lowest aggression rates ([11]; see also column ‘Aggression rates’ in **Table 2**). Yet, lactating females directed aggression of greater score than cycling females, despite the fact that they exhibit lower aggression rates than cycling females ([11]; **Table 2**, column ‘Aggression rates’). Lactating females have greater proximity to alpha males [56, 57], potentially offering them greater access to resources with no need for direct female-female competition (low aggression rates), but it might also buffer female-female aggression risk if males mediate/intervene in female aggressive interactions [15, 29], allowing lactating females to direct aggression to generally more powerful rivals (high aggression score), even if this aggression is infrequent.

Our study suggests that aggression heuristics depend on the social environment and individual needs, meaning that the variation in heuristics observed within or between species [8] at least partially reflects differences in the conditions that specific animal populations or individuals experience at different time points. Our study also adds to behavioural observations from several species suggesting that individuals with greater needs might engage in more risky behaviors [58–60], including inter-individual competition [61–63]. Accordingly, it improves our understanding on the evolution of risk-taking in hominids, including (aggressive) competition in humans where individuals unsuccessful in economic or mating competition may exhibit risky aggressive behaviours [64–68]. Finally, our results may provide some insights regarding the evolution of more egalitarian or despotic societies of other species: if certain social factors or individual conditions can drive shifts in aggression up or down the hierarchy, then they have the potential to flatten or reinforce the hierarchy, respectively.

Data availability

The data and code necessary to replicate this study are available at <https://gitlab.com/nksmt/GorillaHeuristics>.

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Additional information

Ethics

We followed the regulations of Agence Nationale des Parcs Nationaux and the Centre National de la Recherche Scientifique et Technique of Gabon as well as the regulations of Uganda Wildlife Authority and the Uganda National Council of Science and Technology in Uganda. Ethical clearance was given by the Max Planck Society.

Authors' contributions

N.S.: conceptualization, data curation, formal analysis, investigation, methodology, writing—original draft and writing—review and editing; M.M.R.: data curation, funding acquisition, investigation, methodology, writing—review and editing.

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Joint Public Review:

Summary:

This work aims to improve our understanding of the factors that influence female-on-female aggressive interactions in gorilla social hierarchies, using 25 years of behavioural data from five wild groups of two gorilla species. Researchers analysed aggressive interactions between 31 adult females, using behavioural observations and dominance hierarchies inferred through Elo-rating methods. Aggression intensity (mild, moderate, severe) and direction (measured as the rank difference between aggressor and recipient) were used as key variables. A linear mixed-effects model was applied to evaluate how aggression direction varied with reproductive state (cycling, trimester-specific pregnancy, or lactation) and sex composition of the group. This study highlights the direction of aggressive interactions between females, with most interactions being directed from higher- to lower-ranking adult females close in social rank. However, the results show that 42% of these interactions are directed from lower- to higher-ranking females. Particularly, lactating and pregnant females targeted higher-ranking individuals, which the authors suggest might be due to higher energetic needs, which increase risk-taking in lactating and pregnant females. Sex composition within the group also influenced which individuals were targeted. The authors suggest that male presence buffers female-on-female aggression, allowing females to target higher-ranking females than themselves. In contrast, females targeted lower-ranking females than themselves in groups with a larger ratio of females, which supposes a lower risk for the females since the pool of competitors is larger. The findings provide an important insight into aggression heuristics in primate social systems and the social and individual factors that

influence these interactions, providing a deeper understanding of the evolutionary pressures that shape risk-taking, dominance maintenance, and the flexibility of social strategies in group-living species.

The authors achieved their aim by demonstrating that aggression direction in female gorillas is influenced by factors such as reproductive condition and social context, and their results support the broader claim that aggression heuristics are flexible. However, some specific interpretations require further support. Despite this, the study makes a valuable contribution to the field of behavioural ecology by reframing how we think about intra-sexual competition and social rank maintenance in primates.

Strengths:

One of the study's major strengths is the use of an extensive dataset that compiles 25 years of behavioural data and 6871 aggressive interactions between 31 adult females in five social groups, which allows for a robust statistical analysis. This study uses a novel approach to the study of aggression in social groups by including factors such as the direction and intensity of aggressive interactions, which offers a comprehensive understanding of these complex social dynamics. In addition, this study incorporates ecological and physiological factors such as the reproductive state of the females and the sex composition of the group, which allows an integrative perspective on aggression within the broader context of body condition and social environment. The authors successfully integrate their results into broader evolutionary and ecological frameworks, enriching discussions around social hierarchies and risk sensitivity in primates and other animals.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Recommendations for the authors):

Suggestions:

Although this study has an impressive dataset, I felt that some parts of the discussion would benefit from further explanation, specifically when discussing the differences in female aggression direction between groups with different sex compositions. In the discussion is suggested that males buffer female-on-female aggression and that they 'support' lower-ranking females (see line 212), however, the study only tested the sex composition of the group and does not provide any evidence of this buffering. Thus, I would suggest adding more information on how this buffering or protection from males might manifest (for example, listing male behaviours that might showcase this protection) or referencing other studies that support this claim. Another example of this can be found in lines 223-224, which suggests that females choose lower-ranking individuals when they are presented with a larger pool of competitors; however, in lines 227-228, it's stated that this result contradicts previous work in baboons, which makes the previous claim seem unjustified. I recommend adding other examples from studies that support the results of this paper and adding a line that addresses reasons why these differences between gorillas and baboons might be caused (for example, different social dynamics or ecological constraints). In addition, I suggest the inclusion of physiological data such as direct measures of energy expenditure, caloric intake, or hormone levels, as it would strengthen the claims made in the second paragraph of the discussion. However, I understand this might not be possible due to data or time constraints, so I suggest adding more robust justification on why lactation and pregnancy were used as a

proxy for energetic need. In the methods (lines 127-128), it is unclear which phase of the pregnancy or lactation is more energetically demanding. I would also suggest adding a comment on the limitations of using reproductive state to infer energetic need. Lastly, if the data is available, I believe it would be interesting to add body size and age of the females or the size difference between aggressor and target as explanatory variables in the models to test if physiological characteristics influence female-on-female aggression.

Male support:

We have now added more references (Watts 1994, 1997) and enriched our arguments regarding male presence buffering aggression. Previous research suggests that male gorillas may support lower-ranking females and they may intervene in female-female conflicts (Sicotte 2002). Unfortunately, our dataset did not allow us to test for male protection. We conduct proximity scans every 10 minutes and these scans are not associated to each interaction, meaning that we cannot reliably test if proximity to a male influences the likelihood to receive aggression.

Number of competitors and choice of weaker competitors:

We added a very relevant reference in humans, showing that people choose weaker competitors when they have they can choose. We removed the example to baboons because it used sex ratio and the relevance to our study was not that straightforward.

Reproductive state as a proxy for energetic needs:

We now mention clearly that reproductive state is an indirect measure of energetic needs.

We rephrased our methods to: "Lactation is often considered more energetically demanding than pregnancy as a whole but the latest stages of pregnancy are highly energetically demanding, potentially even more than lactation"

Unfortunately, we do not have access to physiological and body size data. Regarding female age, for many females, ages are estimates with errors up to a decade, and thus, we choose not to use them as a reliable predictor. Having accurate values for all these variables, would indeed be very valuable and improve the predicting power of our study.

Recommendations for writing and presentation:

Overall, the manuscript is well-organised and well-written, but there are certain areas that could improve in clarity. In the introduction, I believe that the term 'aggression heuristic' should be introduced earlier and properly defined in order to accommodate a broader audience. The main question and aims of the study are not stated clearly in the last paragraph of the introduction. In the methods, I think it would improve the clarity to add a table for the classification of each type of agonistic interactions instead of naming them in the text. For example, a table that showcase the three intensity categories (severe, mild and moderate), then then dives into each behaviour (e.g. hit, bite, attack, etc.) and a short description of these behaviours, I think this would be helpful since some of the behaviours mentioned can be confusing (what's the difference between attack, hit and fight?). In addition, in line 104, it states that all interactions were assigned equal intensity, which needs to be explained.

We now define aggression heuristics in both the abstract and the first paragraph of the introduction. We have also explained aggressive interactions that their nature was not obvious from their names. Hopefully, these explanations make clear the differences among the recorded behaviours.

We have now specified that the “equal intensity” refers to avoidances and displacements used to infer power relationships: “We assigned to all avoidance/displacement interactions equal intensity, that is, equal influence to the power relationship of the interacting individuals”

Minor corrections:

(1) In line 41, there is a 1 after 'similar'. I am unsure if it's a mistake or a reference.

We corrected the typo.

(2) In lines 68-69, there is mention of other studies, but no references are provided.

We added citations as suggested.

(3) Remove the reference to Figure 1 (line 82) from the introduction; the figure should be referenced in the text just before the image, however, your figure is in a different section.

We removed the reference as suggested.

(4) Line 98 and 136, it's written 'ad libitum' but the correct spelling is 'ad libitum'.

We corrected the typo.

(5) Figure 3, remove the underscores between the words in the axis titles.

We removed the underscores.

Reviewer #2 (Recommendations for the authors):

Here, I have outlined some specific suggestions that require attention. Addressing these comments will enhance the readability and enhance the quality of the manuscript.

(1) L69. Add citation here, indicating the studies focusing on aggression rates.

We added citations as suggested.

(2) L88. The study periods used in this study and the authors' previous study (Reference 11) are different. So please add one table as Table 1 showing the details info on the sampling efforts and data included in their analysis of this study. For example, the study period, the numbers of females and males, sampling hours, the number of avoidance/displacement behaviors used to calculate individual Elo-ratings, and the number of mild/moderate/severe aggressive interactions, etc.

We have now added another table, as suggested (new Table 1) and we have also made clear that we used the hierarchies presented in detail in (Smit & Robbins 2025).

(3) L103. If readers do not look over Reference 25 on purpose, they do not know what the authors want to talk about and why they mention the optimized Elo-rating method. Clarify this statement and add more content explaining the differences between the two methods, or just remove it.

We rephrased the text and in response to the previous comment, we clearly state that there are more details about our approach in Smit & Robbins 2025. At the end of the relevant

sentence, we added the following parenthesis “(see “traditional Elo rating method”; we do not use the “optimized Elo rating method” as it yields similar results and it is not widely used)” and we removed the sentence referring to the optimized Elo-rating method.

(4) L110. Here, the authors stated that the individual with the standardized Elo-score 1 was the highest-ranking. L117, the "aggression direction" score of each aggressive interaction was the standardized Elo-score of the aggressor, subtracting that of the recipient. So, when the "aggression direction" score was 1, it should mean that the aggressor was the highest-ranking and the recipient was the lowest-ranking female. This is not as the authors stated in L117-120 (where the description was incorrectly reversed). Please clarify.

The highest ranking individual has indeed Elo_score equal to 1 and we calculated the interaction score (or "aggression direction score") of each aggressive interaction by subtracting the standardized Elo-score of the aggressor from that of the recipient (Elo_recipient – Elo_aggressor). So, when the aggressor is the lowest-ranking female (Elo_score=0) and the recipient the highest-ranking female one (Elo_score=1), the "aggression direction score" is $1 - 0 = 1$.

(5) Regarding point 3 of the Public Review, please also revise/expand the paragraph L193-208 in the Discussion section accordingly.

Please see our response to the public review. We have enriched the results section, added pairwise comparisons in a new table (Table 2) and modified the discussion accordingly.

(6) Table 1. It's not clear why authors added the column 'Aggression Rate' but did not provide any explanation in the Methods/Results section. How did they calculate the correlation between each tested variable and the "overall adult female aggression rates"? Correlating the number of females in the first trimester of female pregnancy with the female aggression rates in each study group? What did the correlation coefficients mean? L202-204 may provide some hints as to why the authors introduced the Aggression Rate. But it should be made clear in the previous text.

We now added more details in the legend of the table to make our point clear: “To highlight that aggression rates can increase due to increase in interactions of different score, we also include the effect of some of the tested variables on overall adult female aggression rates, based on results of linear mixed effects models from (Smit & Robbins 2024).” We did not include detailed methods to calculate those results because they are detailed in (Smit & Robbins 2024). We find it valuable to show the results of both aggression rates and aggression directionality according to the same predictor variables as a means to clarify that aggression rates and aggression directionality are not always coordinated to one another (they do not always change in a consistent manner relative to one another).

(7) L166. This is not rigorous. Please rephrase. There is only one western gorilla group containing only one resident male included in the analysis.

We have toned down our text: “Our results did not show any significant difference between female-female aggression patterns within the one western and four mountain gorillas groups”

(8) L167. I don't think the interaction scores in the third trimester of female pregnancy were significantly higher than those in the first trimester. The same concern applies in L194-195.

We have now added a new table with post hoc pairwise comparisons among the different reproductive states that clarifies that.

| (9) L202. *There is no column 'Aggression rates' in Table 1 of Reference 11.*

We have rephrased to make clear that we refer to Table 1 of the present study.

| (10) L204-205. *Reference 49. Maybe not a proper citation here. This claim requires stronger evidence or further justification. Additionally, please rephrase and clarify the arguments in L204208 for better readability and precision.*

We have added three more references and rephrased to clarify our argument.

| **Reviewer #3 (Recommendations for the authors):**

| (1) Line 41: *The word "similar" is misspelled.*

We corrected the typo.

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